

Assessment of Camera Capture for GPS RFI Monitor

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Abstract—Today, the global positioning system (GPS) has become the enabling technology of the communication, navigation, surveillance, and air traffic management (CNS/ATM) systems. However, the transmitting power of a GPS signal is extremely low and thus vulnerable to intentional or unintentional radio frequency interference (RFI). Therefore, this paper proposes a terrestrial GPS RFI monitor to identify a possible RFI source to reduce the impact on aviation operation. Each GPS RFI monitor includes a software receiver with either an automatic gain control (AGC) based or a modified energy based RFI detector in order to detect possible interference. The GPS RFI monitor also includes a camera to record the corresponding video stream simultaneously with a RFI event. If a GPS RFI device is present, the RFI monitor and camera will be triggered. Once the detector has been triggered, the associated intermediate frequency (IF) data and video stream are saved. This makes it possible to identify the characteristics of the possible GPS RFI device and to determine the time as well as the impact of the RFI event. In this paper, the recorded GPS IF signal is first processed to draw the time-frequency figures to evaluate the characteristics of the possible GPS interference, and then processed by a software-defined radio (SDR) receiver to assess the impact on positioning. The results from previous studies are compared with several RFI detectors whose detection schemes are presented in this paper as well.

Keywords—interference; GPS RFI monitor; AGC

I. INTRODUCTION

Today, people worldwide rely on global positioning system (GPS) satellite navigation to deliver position, velocity, and time information [1]. Meanwhile, with the development of augmentation systems and regional satellite systems, the performance of GPS is improved by modeling the error sources. However, the transmitting power of a GPS signal is quite low and thus vulnerable to intentional or unintentional radio frequency interference (RFI). Any signal that is close to the GPS L1 band might be able to pass the bandpass filter of one GPS frontend, and the noise floor will be therefore raised such that the receiver cannot acquire or track the GPS satellites. Even more concerning is that GPS jammers can be obtained easily on the internet at very low cost despite the fact that they are illegal in most countries. Based on previous research, there were a lot of cases that interference to GPS was caused by

jammers carried by vehicles. Because GPS is the enabling technology of modern civil aviation applications, there would be a significant impact on the aviation operations if GPS service was degraded. Therefore, methods to detect and identify the possible RFI sources to GPS were developed.

The main objective of this paper was to design a GPS RFI monitoring system with some modified energy detectors which are implemented on a software defined radio (SDR) platform called universal software radio peripheral (USRP). The focus was on detection and analysis of GPS RFI sources on moving vehicles in the specific area of interest. Each GPS RFI monitor has a software receiver with either an automatic gain control (AGC) based or a modified energy based RFI detector to detect possible interference. The GPS RFI monitor also contains a camera that records the corresponding video stream of the area during a RFI event. The camera is facing the road and the GPS antenna is oriented sideways toward where the camera is pointing to have better antenna sensitivity in the direction of interest. Ideally, if a vehicle carries a GPS RFI device and drives by the road, the RFI monitor will be triggered which in turn triggers the camera to record video. Once the detector is triggered, the associated intermediate frequency (IF) data is saved. This makes it possible to identify the vehicle carrying a GPS RFI device and to determine the time as well as the impact of the RFI event. In addition, the recorded GPS IF signal is processed to draw the signal energy distribution and time-frequency figures in order to evaluate the characteristics of the possible GPS interference with the recorded video streams. Finally, a SDR receiver is used to process the recorded GPS IF signal to assess the impact on positioning performance.

Accordingly, this paper is organized as follows. Section II describes the detection algorithms for RFI to GPS and introduces the method being implemented in this paper. In Section III, the design of the detector and the methods to determine the presence of RFI are discussed. Section IV presents the experiment setup as well as the experiment results. Finally, the final section includes the conclusion and some suggestions for the future work.

II. GPS RFI DETECTION AND ANALYSIS

In the past few years, several GPS interference detection algorithms had been developed [3-6]. This section discusses several commonly used GPS RFI detection algorithms and the terrestrial GPS RFI monitor proposed in this paper.

A. Automatic Gain Control Measurement

Generally, most receiver frontends include automatic gain control (AGC) to control the dynamic range of the input signal to the receiver amplifier. AGC detects the input signal level, compares it with a reference value, utilizes the difference to control variable gain amplifiers (VGA), and adjusts the magnitude of the input signal to protect the receiver from being damaged by receiving an excessively strong signal. Therefore, the AGC level measures the average value of the input signal and adjusts the amplifier gain in order to keep the output signal constant. This means an abnormal AGC level recorded by the GPS receiver could be regarded as an indication for interference. Fig. 1 shows the distribution of AGC level from a NordNav GPS receiver frontend. If there was a RFI event, the AGC value would decrease. In this way, the AGC level can be a very powerful detector after setting up an appropriate threshold. Some applications of an AGC level detector were proposed in [5-7].

However, the USRP used in this work cannot output AGC values. Fortunately, there are numerous interference detectors that are based on various properties of the received signal. One of the established detection methods is the energy detection measurement. The energy detection method is used in many applications and it is introduced in the next section.

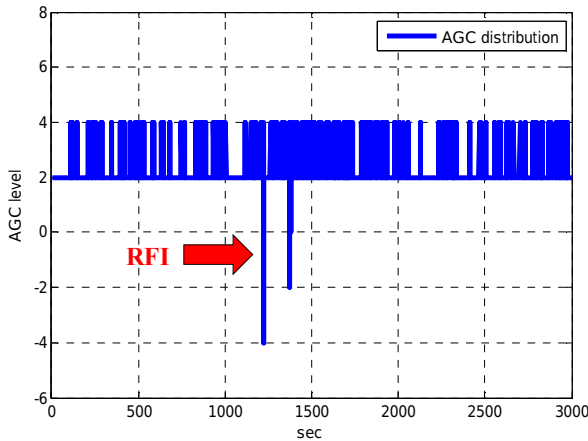


Fig. 1. The AGC level changes when an RFI appears.

B. Energy Detector Measurement

An energy detector measures the received signal energy for a finite time space and compares it to a predefined decision threshold that can be used to detect many kinds of jamming interference [8], but only at static state. Variations of the standard energy detector can be performed in the frequency domain and the time domain. In the frequency domain, a detector using an averaged fast Fourier transform (FFT) and power spectral density (PSD) of standard periodogram was

proposed in [9] for detecting a narrowband continuous wave interference (CWI). A similar approach was also taken in [5] and [10], but it used the improved periodogram to find the characteristics of the interference signal, which added a window function to correct the drawback of the previous method, and thus was called a Welch periodogram. For example, as shown in Fig. 2, the features of the time-frequency plot produced by a Welch periodogram such as the frequency band, event time and signal power can be clearly identified, and the strong signal nearby the center frequency of GPS L1 band indicated by the red arrow is the possible interference. A Welch periodogram has better performance than that of a standard periodogram for indicating the characteristics of a signal. The differences between the Welch periodogram and the standard one can be found in [11]. Therefore, the system developed in this paper uses a Welch periodogram to plot time-frequency figures for analyzing the RFI event.

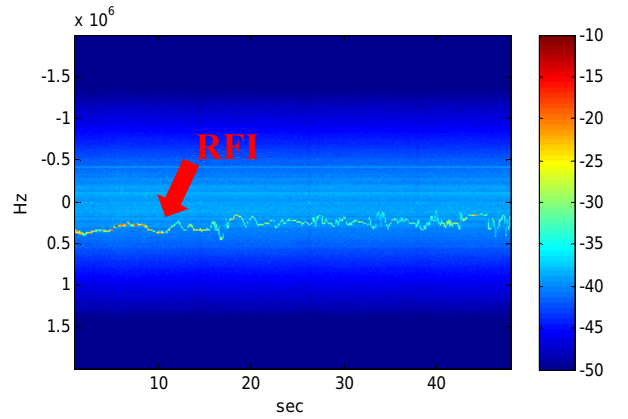


Fig. 2. Time-frequency plot by Welch periodogram.

Another option for an RFI detector is to use the Carrier-to-Noise (C/N_0) ratio in time domain [12]. C/N_0 is a common measurement of the signal quality from a GPS satellite at the receiver. Although C/N_0 can be used to detect the interference, it cannot distinguish between the increased jamming signal power and the decreased GPS signal power so that it might lead to false alarms [13]. For example, the C/N_0 estimates would decrease during an RFI event or when the GPS signal is weakened due to blockage by a building or other objects. The RFI detector proposed in [14] successfully applied the C/N_0 to overcome the above drawback by recording C/N_0 at the same place for many days. A characteristic model of C/N_0 was then applied to detect a GPS interference signal.

In this paper, the RFI detector algorithm is very similar to the energy detector measurement method but with some modifications, and is thus called the modified energy detector. The details of the modified energy detector are explained in the following section.

C. Modified Energy Detector Measurement

This paper uses an energy metric based on raw intermediate frequency (IF) data processed from the USRP [13]. The type of raw IF data is complex which is arrayed into a real part (I) and an imaginary part (Q). Raw IF data were collected when the USRP started, and the standard deviation as the indication of signal energy for every unit of time was calculated. For

example, if the sampling frequency is 4 MHz and the buffer is 1 MHz, this method can calculate a point of standard deviation as signal energy indication per 0.25 seconds. In addition, those points of standard deviations are very important and will be referred to as energy distribution in this paper.

Under normal conditions, the magnitude of the GPS signal is at some stable level such that the signal energy distribution is well behaved. When interference affects GPS, fragments of the GPS signal magnitude are increased and the energy distribution is increased at the same time as depicted in Fig 3. This figure shows the energy distribution recorded for one day where possible RFI events occurred at about 7:00 A.M, 8:00 A.M, and 14:00 P.M.

As a result, effectively identifying the existence of RFI using the signal energy distribution is the objective of this paper and the details of the implemented detector design will be shown in the following section.

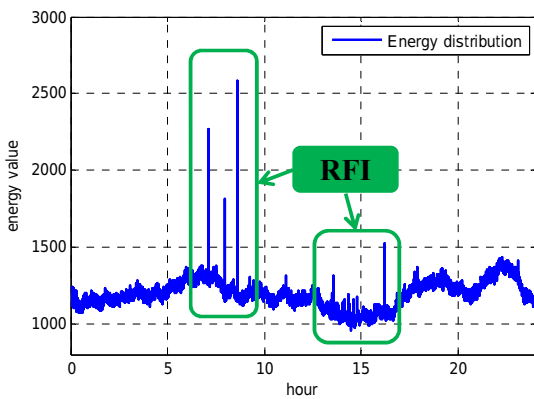


Fig. 3. Signal distribution with modified energy measurement from USRP.

III. DETECTOR DESIGN

Three different detector designs based on signal energy distribution are described in this section. In addition, analysis on detection thresholds and a comparison of different detectors is presented. In order to detect RFI effectively, the detector designs generally focus on finding the strong RFI case first, adjusting the parameters to detect the normal RFI case, and finally detecting the all RFI case. This section first introduces each of the three detector designs, and then chooses the effective detector design to be implemented in this GPS RFI monitor system.

A. The Detector Design 1 (for Strong Interference)

The threshold in this detector is the simplest one of all detector designs. Instead of processing raw IF data directly, the value of energy distribution for every unit of time for a day is recorded. After that, the mean value and the standard deviation (σ) for the recorded energy distribution are calculated. A value of 5σ is used as the threshold and is set in the RFI monitor system for the RFI detection test. For example, the threshold is set as the red line shown in Fig. 4. If the energy distribution (blue line in Fig. 4) exceeds the pre-determined threshold, then it suggests presence of a RFI event, and the RFI recording mechanisms of this system would be triggered at the same time.

The more the value exceeds the threshold, the stronger the RFI is assumed to be. The advantage of this method is simplicity so that it can be implemented quickly.

However, the disadvantage of this detector design is that it can only be used to detect strong RFI because it uses a fixed threshold. Nevertheless, the energy distribution will change over time along with the movement of GPS satellites. In Fig. 4, the experiment shows that the energy distribution at night is higher than that at morning, since there are more GPS satellites in view at night in Taiwan. As a result, the energy distribution at night is closer to the fixed threshold, but the energy distribution in the morning is farther away from the fixed threshold. This means the same RFI event would be more likely to be detected at night than in the morning. On the other hand, the false alarm rate of this detector design will also be higher at night.

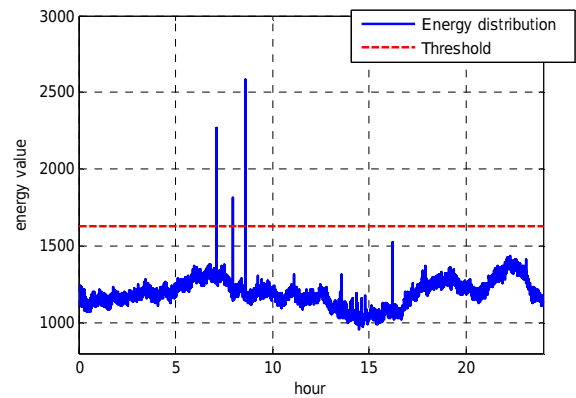


Fig. 4. Strong detector simulation.

B. The Detector Design 2 (for Normal Interference)

In this section, the detector design for normal strength RFI is illustrated. Detector design for normal strength RFI is more complicated than that for strong RFI because it needs more information to build an appropriate threshold database. In general, GPS satellites repeat the same ground track every day. Therefore, GPS signal energy distribution will repeat the same pattern for a static (fixed) GPS receiver, because the environment is basically the same. For instance, the energy distribution shown in Fig. 5 is recorded for 5 days. If the data is shifted by 300 units of time for each day, then the energy distributions are obviously showing some regular pattern every day. However, as shown in Fig. 5, there are a few anomalies in the collected dataset such that it could not be used to determine the RFI detection threshold directly. For this reason, the collected data must be processed to exclude the abnormal data. The polynomial regression algorithm was applied to solve this problem. The average energy distribution value of the 5 days was found, and the new curve was calculated by the polynomial regression algorithm. After doing so, the standard deviation from the curve was computed again, and $3\sigma \sim 5\sigma$ was added to the curve as the threshold database. The simulation result is shown in Fig. 6. The red curve is the threshold database computed by the polynomial regression method, and the blue line represents the actual energy distribution. As indicated in Fig. 6, ignoring those peaks, the threshold database

and energy distribution are similar to each other, but the threshold database is much smoother than the energy distribution. In this way, not only a strong RFI event but also a normal RFI event can be detected. However, the disadvantages of this RFI detector design are that this detector design might have a higher false alarm rate in a complex environment and this detector design needs to record baseline data for an extended period of time to determine the detection threshold.

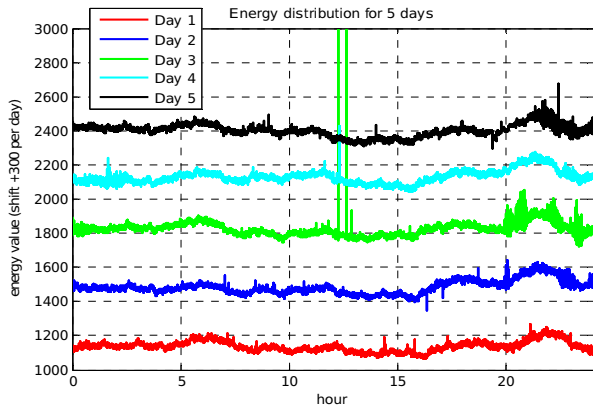


Fig. 5. Energy distribution for five days.

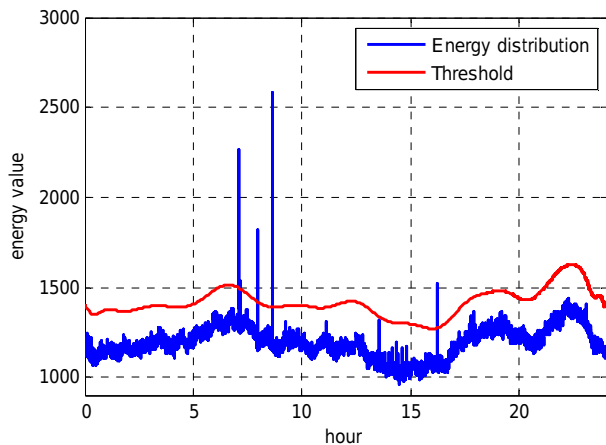


Fig. 6. Normal detector simulations.

C. The Detector 3 (for any Interference)

For fast and real-time implementation of a RFI detector, there is a need to establish the detection threshold in real time. The RFI detector design presented in this section is the most complex among the three detector design presented in this paper. For this RFI detector, the important characteristics of the threshold are that it must be able to adjust itself over time and it should not record energy distribution as the other two methods do. This RFI detector uses the energy distribution for a short period of time and then adds a reference value to be set as the current RFI detection threshold. As depicted in Fig. 7, the threshold is updated over time under nominal conditions. However, if one RFI event occurs, the detection threshold would be locked to the value until 40 seconds after the RFI disappears.

A simulation result is shown in Fig. 8. The energy distribution is shown by the blue line and the threshold is shown by the red line. The threshold is set according to the previous energy distribution after 2 seconds with a constant shift and it updates every second. When the current energy level was higher than the threshold, there would be a cross point as shown by the green circle in the figure. At the same time, this detector would lock the current threshold until 40 seconds after the energy level resumed to normal as shown by the lower plot in Fig. 8.

This detector design is very sensitive to any changes in signal energy, especially in the case that the signal is becoming weak or strong due to the changes in the number of in-view satellites. There are two weaknesses in this detector design that must be mentioned. One weakness is the reference value is added to the threshold, which shifts the threshold and affects the sensitivity of this detector. In this paper, the reference value is set based on empirical experience. The other weakness is the time delay due to the first short period of time while building the threshold. In that period of time, the detector could not detect a RFI event. In addition, the length of time to establish the detection threshold will also affect the sensitivity of the detector. For this reason, this RFI detector might not work properly if it is used for a long period of time, and this detector design might not tolerate the dynamic movement, because a GPS signal may be blocked by buildings or other objects such that the threshold might be set incorrectly.

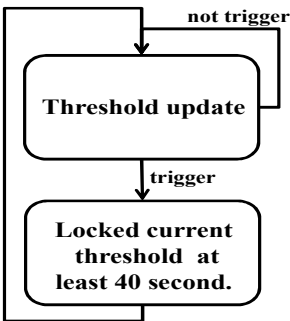


Fig. 7. The flowchart of threshold update in sensitive detector.

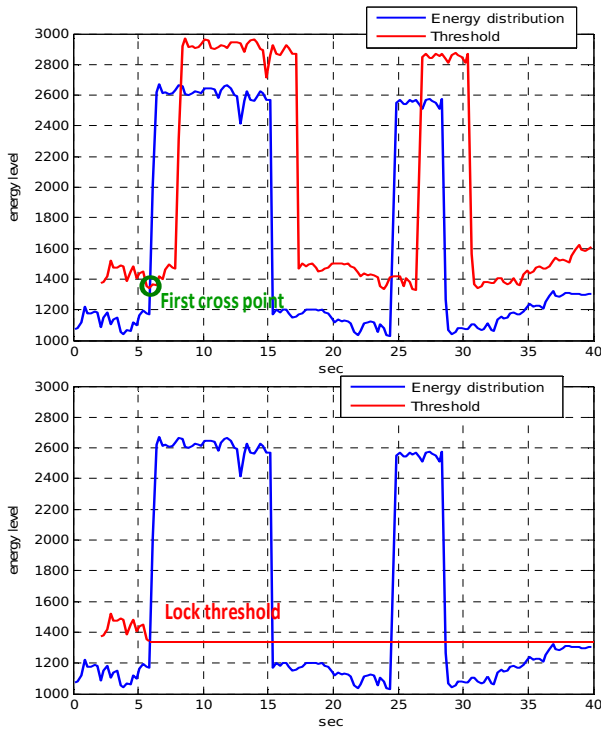


Fig. 8. The simulation of sensitive detector.

In Table I, the three detectors presented in this section are summarized for comparison. Due to the fact that extreme RFI may be the most hazardous, detector 1 was selected to be used in this paper. The results will be discussed in the next section.

Table I The comparison table of three RFI detector designs.

	Detector 1	Detector 2	Detector 3
Complexity	low	medium	high
Sensitivity	low	medium	high
Support for long term operation	yes	yes	no
False alarm rate	low	medium	high
Characteristics of the detector	1. For strong RFI. 2. Can be applied to any place of interest. 3. Threshold is fixed.	1. For strong and normal RFI. 2. Can be applied to any place of interest after building a database. 3. Threshold is determined by database.	1. For any kind of RFI. 2. Can be applied to any place of interest in very short time. 3. Threshold is updated every unit of time.

IV. EXPERIMENT SETUP AND RESULT

This section illustrates the GPS RFI experiment setup and results of the experiments that were conducted at the National Cheng Kung University in Taiwan. First, the hardware and software architecture of the developed GPS RFI monitor system will be introduced. Then, the system flowchart will be described. Finally, several experiment results are presented.

A. Hardware and Software Architecture of RFI Monitor

In this work, the GPS RFI monitor system used an USRP N210 as the frontend. With an Ethernet interface it is possible to record GPS IF data. In addition, the hardware of the system also contained a laptop with a Linux operating system, a NovAtel 703GGG antenna, and a camera, as illustrated in Fig. 9. In terms of software, QT was used to build a graphical user interface (GUI) to control the USRP N210 frontend and the camera in real time. MATLAB was then used as the data analysis tool.

A laptop runs the GUI to record the IF data from the USRP N210 and to capture the video stream from camera. When the system is operational, it captures a video frame per second and displays it inside the GUI. There were six canvases on the interface for saving six video frames before the RFI event occurred. In this way, when a RFI event occurred, the camera would be triggered to capture not only the six video frames before the trigger but also at least one video frame per second until the interference disappeared. At the same time that the RFI event occurs, the RFI monitor system will save the previous 40 seconds of IF data for analysis.

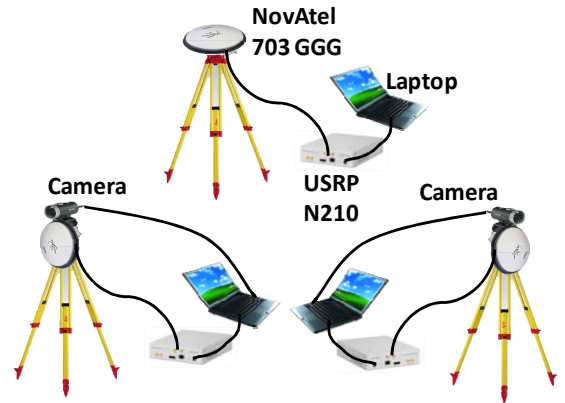


Fig. 9. Hardware setup of a GPS RFI monitors.

B. GPS RFI Monitor System Flowchart

The system flowchart of the developed GPS RFI monitor system is depicted in Fig. 10. As shown in Fig. 10, the solid lines and frames represent the real time processes, and the dashed lines and frames indicate the post-process analyses.

In the beginning of the flowchart, a GPS signal is received by the GPS antenna, and the signal is passed through the low-noise amplifier (LNA) to perform signal conditioning. Then, the RF signal is converted into a raw IF signal by down-conversion and a RF filter. It is then passed to the AGC to adjust its level. Finally, the IF signal is transformed into a digital signal by the analog-to-digital converter (ADC). In Fig. 10, the steps in the yellow region are standard signal processes in a general GPS frontend.

After the GPS frontend, the digital signal (raw IF data) is fed into the data processing computer. The raw IF data is processed and computed as the energy value to be detected (as shown in red block in Fig. 10). If an RFI event is occurring, the camera saves the video streams before and after the triggered

event (as shown in purple block in Fig. 10). Also, the GPS RFI monitor system saves the IF data for the time-frequency analysis and user positioning evaluation. In time-frequency analysis (as shown in green block in Fig. 10), the Welch periodogram is applied to plot the time-frequency figure and to investigate the characteristics of the RFI event. On the other hand, a GPS SDR receiver is used to process the recorded IF data to perform the acquisition and tracking processes on the GPS satellite signal, decode the information of the navigation message, and decode the pseudo range measurement. Finally, the GPS satellite position and the user position are computed. Based on the analysis, we can evaluate the RFI impact on the user positioning performance.

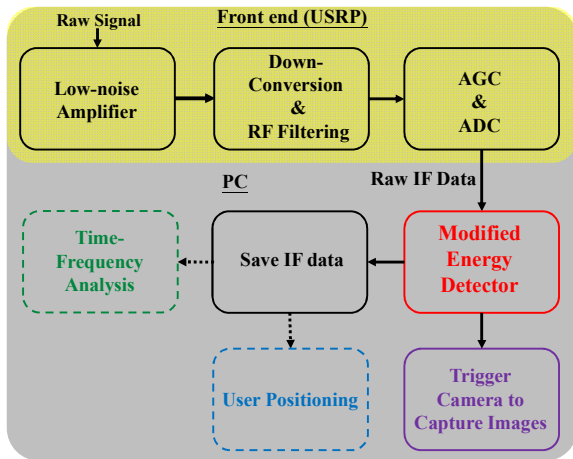


Fig. 10. GPS RFI monitor system flowchart.

C. Experiment setup:

Fig. 11 shows the experimental location and the hardware setup. In Fig. 11, the part of the road of interest is enclosed by the yellow frame. For each of the two stations with a camera (as shown in Fig. 9), the camera is facing the road and the GPS antenna is oriented sideways toward where the camera is pointing, such that the antenna has increased sensitivity in the direction of interest. The received IF data from the roadside stations could be blocked by the trees or other obstacles. Both of them could not receive enough signals from GPS satellites to calculate the user position. For this reason, the third station, which is the station without a camera, was chosen to be set up on the top of an antenna tower, pointing the antenna to the sky. The received IF signal from top station was able to generate a position solution. The positions of the stations are shown in Fig. 11.

In addition to the experiment setup, two points must be discussed before the next section. First, the threshold settings from the roadside stations and the top station were different because of the different environment and antenna settings. Second, the antennas were fixed with a chuck ring to decrease the effect of multipath. The next section will show two simulations about these two points, and then show the real experiment results from the camera and the analysis of the detected RFI events.



Fig. 11. System location and setup.

D. Experiment Results:

In order to increase the reliability of camera results, two simulations for the roadside stations were conducted with the antennas shown in Fig. 11. For the first simulation, an electronic key for a vehicle, which can affect the GPS L1 band, was used to test how the received signal strength varied with different reception angles of a signal from a fixed distance to the antenna. To distinguish different angles clearly in Fig. 12, the electronic lock transmitted signal continuously every 5 seconds. During this, the antenna was rotated by 45 degrees every 5 seconds. The simulation result is shown in Fig. 12. In Fig. 12, 0° marks the situation that the signal comes from directly in front of the antenna; while 90° and 180° indicates that the signal comes from the side and from behind the antenna respectively. As can be seen in the result, 0° angle of reception results in the largest energy value and 90° angle of reception results in the smallest energy value because of the chuck ring. The brief conclusion from this simulation is that a vehicle in front of the camera as the system is triggered may very likely be the origin of RFI.

In the second simulation, the electronic key continuously transmits signal while it is moving toward the user along the road side. In this way, the electronic lock would trigger the roadside stations at different times. The energy distributions of them are shown in Fig. 13 and Fig. 14. In Fig. 13, the electronic lock moved from left to right. Due to the fact that the left antenna faced directly towards the electronic lock, the energy level increased as the electronic key moved towards the antenna. However, when passing the left antenna, the energy distribution on left station (blue line) dropped because of the chuck ring. Almost at the same time, the energy distribution on right station exceeded the predefined threshold and was triggered. When walking far from both stations,

energy distribution on both stations decreased. During that period, the electronic lock faced toward the right station but was behind the left station, which made the energy distribution (green line) decrease more slowly than the other one (blue line). The same result, as shown in Fig. 14, is obtained while walking in the opposite direction. Therefore, this simulation shows that the moving direction of interference can be determined by observing the trend of weakening energy.

With these previous simulation results, the actual experiment results are shown as the follows.

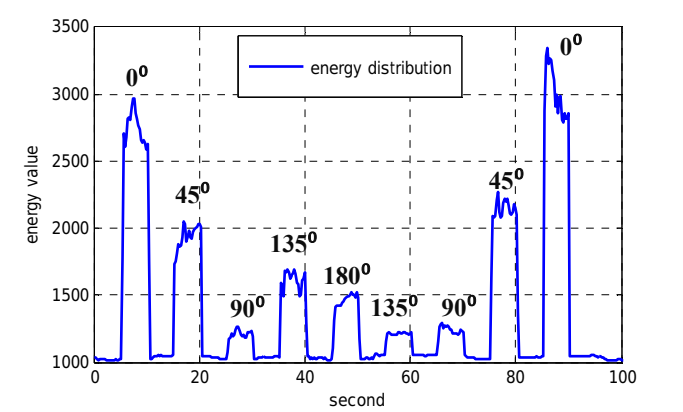


Fig. 12. Simulation for testing angle of antenna.

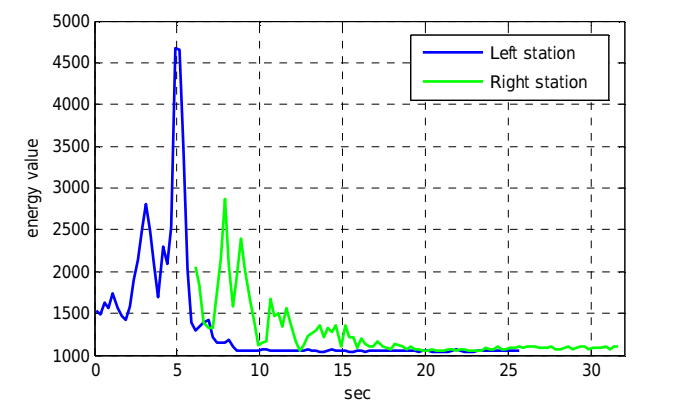


Fig. 13. Simulation for walking from left to right with the electronic lock.

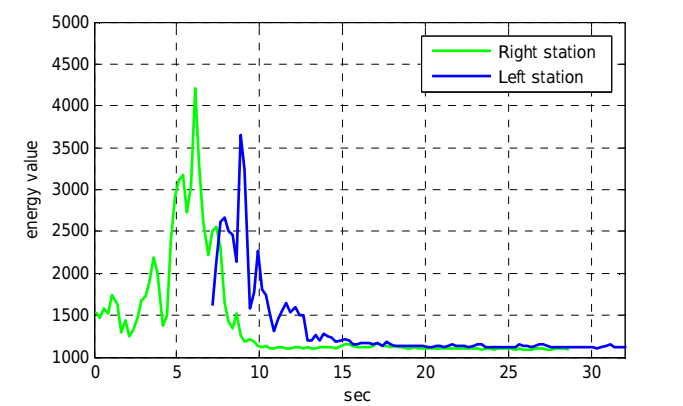


Fig. 14. Simulation for walking from right to left with the electronic lock.

During the experiment, two RFI events occurred on April 11, 2014. In Fig. 15, the blue line and green line respectively

represent the energy distribution on the left station and the right station, and the red line represents energy distribution on the top station. It should be noted that different environments and antenna settings resulted in different thresholds on the roadside stations and the top station. In Fig. 15, the RFI event I sequentially triggered the left station, then the top station and the last right station. The blue peak occurred at the tenth second which means the RFI source may have been in front of the left station. After about two seconds, the blue peak dropped quickly and the green peak appeared at about the twelfth second. Therefore, it can be inferred that the RFI source moved from the left to the right direction. Further information about RFI event I can be drawn by comparing Fig.15 and the snap from the camera shown Fig. 16. A car, which is likely to be the RFI source, is circled by the red dashed line entered the view at about the tenth sec which corresponds to the blue peak in Fig. 15. After about two seconds, the blue peak dropped quickly but the green peak appeared. At the same time, this car moved into view of the right camera. The green line in Fig. 15 decreased more slowly than blue line is as the prior simulation result. Therefore, it is highly likely that the car in Fig. 16 was an RFI source.

The influence of RFI event II lasts about 50 seconds as shown in Fig. 17. In this period, the first peak appeared at about the nineteenth second. Then the energy distribution (blue line and green line) started to drop after about the fifty-eighth second. Meanwhile, a vehicle was focused from the right camera lens at about the nineteenth sec to the left camera lens at about the fifty-eighth sec. The image information from the left camera shows that all of the cars were stopped by the red light during that period (about 40 seconds). A white car circled by red dashed line in Fig. 18 that did not emerge on the left camera lens until the fifty-eighth sec, appears at the right camera lens at about the nineteenth sec. The reasonable explanation for long time influence is that the white car parked at the roadside which exceeded the range of both cameras unexpectedly. Then, the energy distribution of both roadside stations started to decline at about the fifty-eighth sec in Fig. 17. At the same time, the white car entered the view of the left camera as shown in Fig. 18. As the white car moved away, the green line quickly returned to the normal range because the car was behind the right antenna. The blue line gradually resumed to the normal range because the car was facing towards the left antenna.

Without checking the GPS signal position solution, it can still be inferred that these cars may have installed RFI devices. The IF data received from top station was used to analyze the position solution in this RFI case. A narrow CWI occurred at GPS L1 band can be found clearly in Fig. 19. However, this CWI was not powerful enough to affect the user position solution and it will not be shown in this paper.

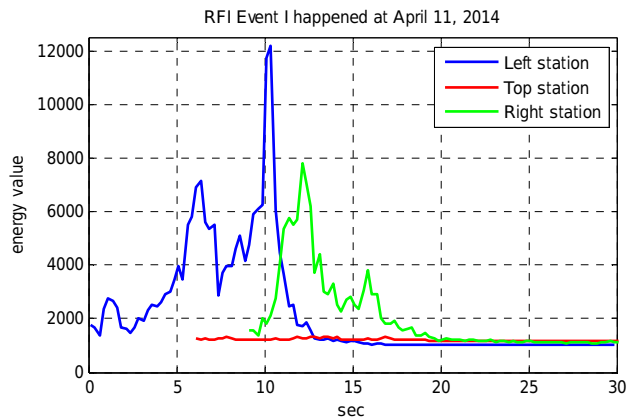


Fig. 15. Energy distribution of RFI event I happened at April 11, 2014.

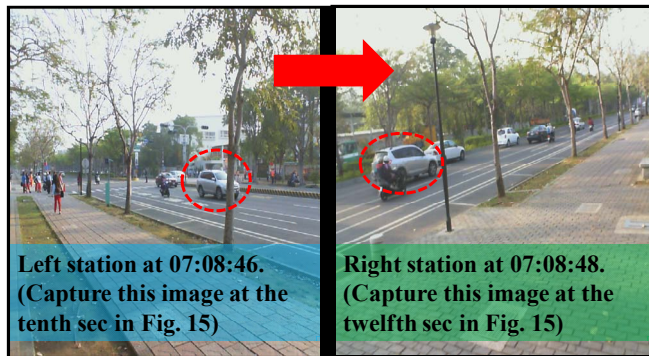


Fig. 16. Possible source of the RFI event I.

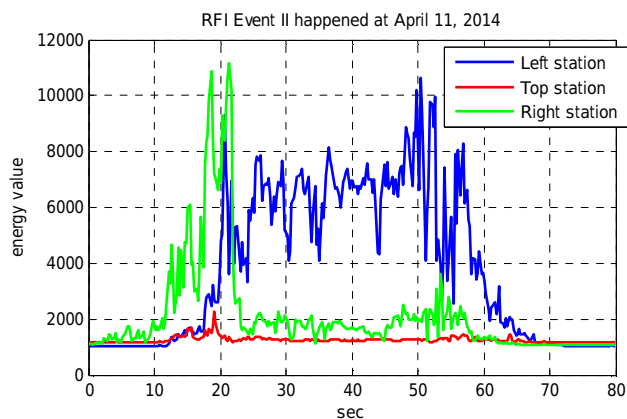


Fig. 17. Energy distribution of RFI event II happened at April 11, 2014.

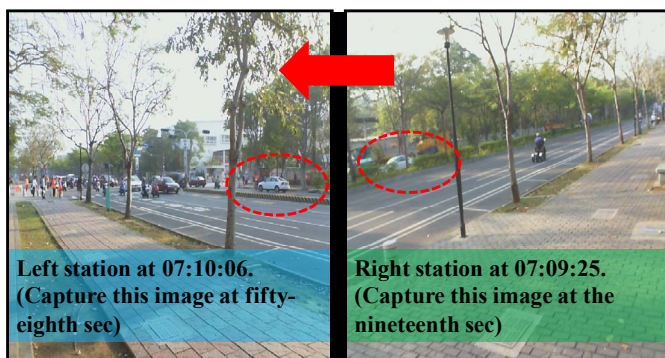


Fig. 18. Possible source of the RFI event II.

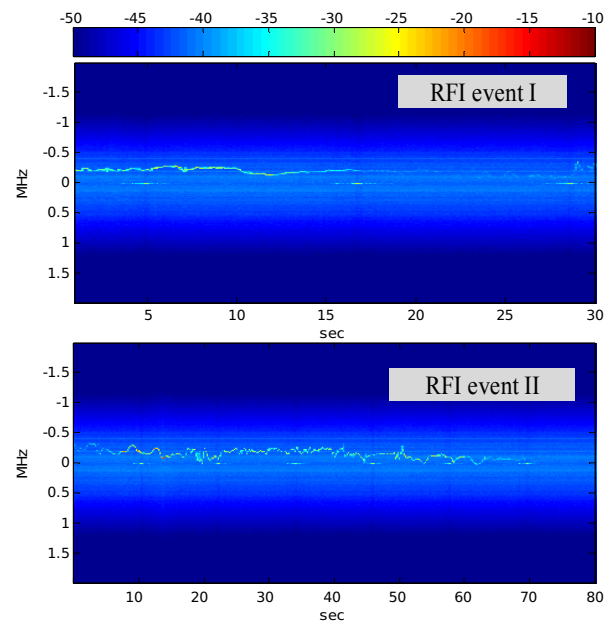


Fig. 19. Time-frequency plot of the RFI case happened at April 11, 2014.

In order to confirm the impact of RFI on user positioning, Fig. 20 to Fig. 23 show the results obtained from another serious RFI event that occurred on March 13, 2014. This serious case was recorded by the top station. Due to the fact that the serious RFI source might be located on other roads than those in the range of camera it was unable to be evaluated by the camera on roadside stations, Fig. 20 shows the time-frequency plot about this serious RFI case; a strong broadband CWI occurred at the first sec to the tenth second. During that period, energy distribution (green line) is abnormally high and C/N_0 estimation (blue line) on all satellites dropped down visibly in Fig. 21. Fig. 22 shows the resulting energy distribution and acquisition of GPS satellites per second. In Fig. 22, the blue points represent successful acquisition; red points represent failure to acquire satellite signal continuously. It can be seen in Fig. 22 that when this strong broadband CWI occurs, it blocked off all GPS satellite signals during that period. On the other hand, decoding navigation information and pseudo range requires at least 4 36-second continuous signals from different satellites. This broadband CWI disallows the SDR to locate the user position until it disappears. The error associated with SDR positioning result is shown in Fig. 23. In Fig. 23 the value -1 indicating failure to locate user position due to the broadband CWI.

To conclude this section, a briefly summary is provided. Detecting RFI events strong enough to affect users positioning results is not easy in a short period experiment. However, this RFI monitor system can effectively verify the presence of RFI event, evaluate the degree of RFI effect on user position, and even identify the possible source of RFI.

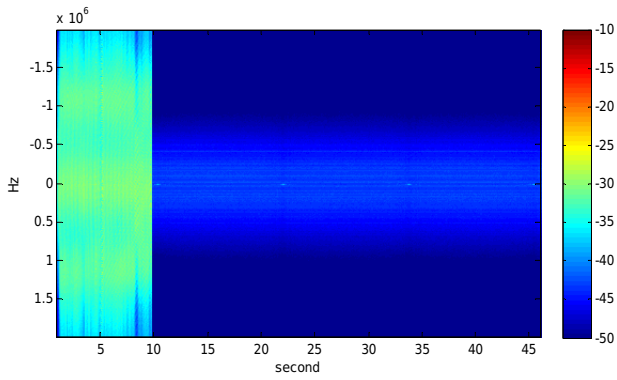


Fig. 20. Time-frequency plot of the serious RFI case happened at March 13, 2014.

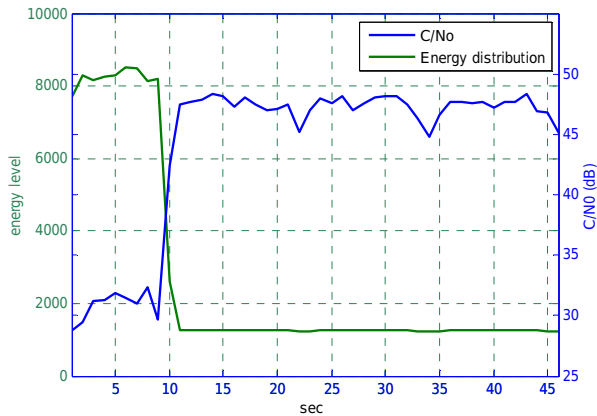


Fig. 21. Energy and C/N_0 distribution of the serious RFI case happened at March 13, 2014.

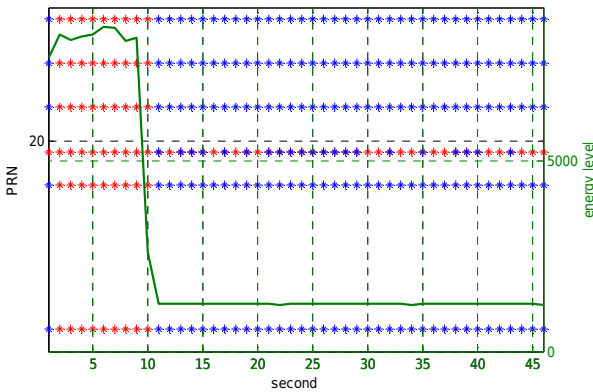


Fig. 22. Energy distribution and GPS signal acquisition figure of the serious RFI case happened at March 13, 2014.

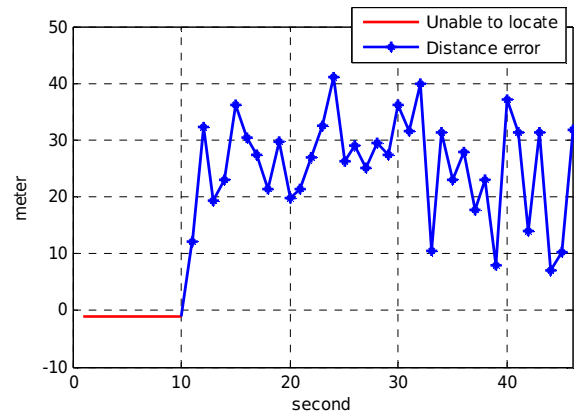


Fig. 23. Distance error figure of the serious RFI case happened at March 13, 2014.

V. CONCLUSION AND FUTURE WORK

This paper presents a GPS RFI monitor system with the modified energy measurement detection algorithm and compares the advantages and disadvantages for each method. Also, two simulations on the antenna angle relative to a RFI source to determine possible RFI sources were discussed.

The experimental results show that the GPS RFI monitor system not only detects GPS L1 interference but also analyzes its influence on the position solution. In addition, the design of a RFI monitor with a camera attached to locate the potential source of the interference is discussed. This setup will be used to detect GPS jammers in cars on the road near the GPS RFI monitor. Correlating the recorded pictures and IF data allows for a car containing a GPS jammer to be located. For future work, more information can be incorporated to increase the confidence on the inference based on the images. The Time Difference of Arrival (TDOA) algorithm to locate the RFI source is a feasible method to achieve this goal. This could increase the confidence in determining the source of RFI.

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